## Q1 Scatterplot: weekly sales vs footfall

**Task 1** *Scatterplot with straight-line fit and loess curve.*

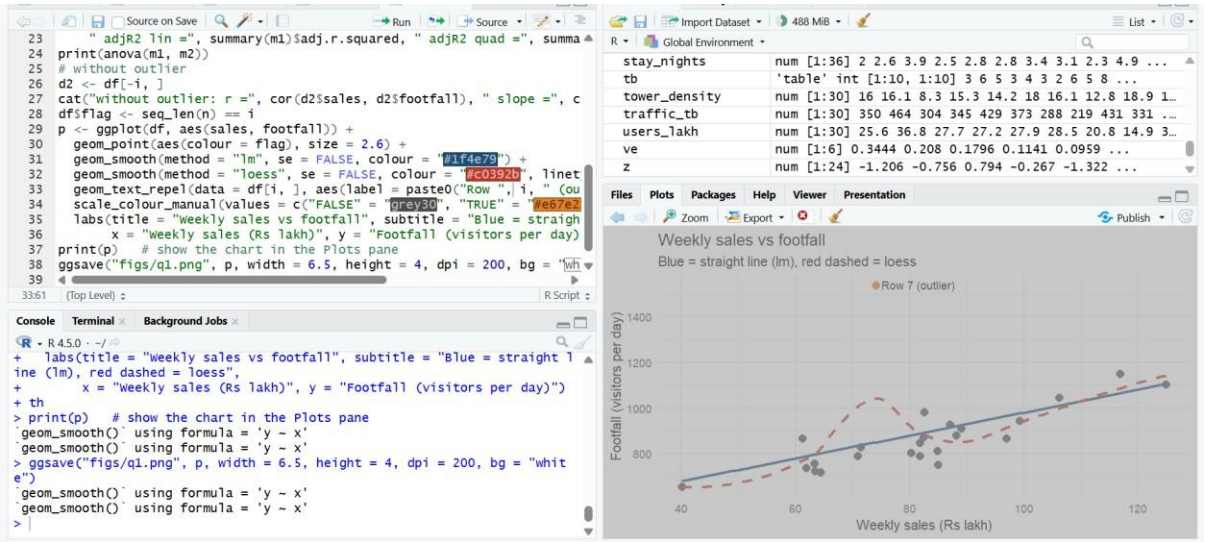



*Figure Q1: Weekly sales (Rs lakh) vs footfall (visitors per day), n = 24. The orange point is row 7.*

**Task 2** *Correlation, slope and one-sentence description.*

**r = 0.51** and **slope = 5.01** visitors per day for every extra Rs 1 lakh of weekly sales (intercept 479.2, $R^2$ = 0.263). The relationship is a **moderate, positive, roughly linear** one: weeks with higher sales tend to have higher footfall, but the fit is loose.

**Task 3** *Flag the largest standardised residual.*

**Row 7** (sales = 74.1 Rs lakh, footfall = 1536 visitors/day) has the largest standardised residual, **rstandard = 4.35**. The next largest is only 0.96, so it stands far apart. The line predicts about 850 visitors for that week, so the record is roughly 686 above the fit, which matches the "keyed in wrongly" note in the code. Without row 7 the correlation rises to **r = 0.87** and the slope to **5.66**.

**Task 4** *Quadratic model vs straight line.*

Straight line $R^2$ = **0.2632**; quadratic $R^2$ = **0.2635**. The quadratic adds almost nothing (adjusted $R^2$ actually falls from 0.230 to 0.193; F-test p = 0.936). The **straight line fits as well as the curve**, so there is no evidence of a non-linear trend. The wiggle in the loess curve is caused by the outlier at row 7.

**Task 5** *Recommendation for the Sales Director.*

First **check and correct the row 7 entry** (1536 looks like a typing error). Once it is fixed, footfall is a reliable guide to sales (r ≈ 0.87): each extra Rs 1 lakh of sales goes with about 5.7 more visitors, so plan staffing and promotions around footfall.

### R code used for Q1

```
library(ggplot2); library(ggrepel)
th <- theme_minimal(base_size = 11)
dir.create("figs", showWarnings = FALSE)

set.seed(1296066)
n <- 24
sales <- round(rnorm(n, 85.0, 18.0), 1)
z <- (sales - 85.0) / 18.0
footfall <- round(900 + 200.0 * (0.56 * z + 0.00 * z^2 + 0.36 * rnorm(n)), 0)
footfall[7] <- round(footfall[7] + 3.6 * 200.0, 0)
df <- data.frame(sales, footfall)
print(summary(df))
m1 <- lm(footfall ~ sales, df)
r1 <- cor(df$sales, df$footfall)
cat("r =", r1, " slope =", coef(m1)[2], " intercept =", coef(m1)[1], " R2 =",
summary(m1)$r.squared, "\n")
rs <- rstandard(m1)
i <- which.max(abs(rs))
cat("max |rstandard| row", i, "sales", df$sales[i], "footfall", df$footfall[i], "rstd",
rs[i], "\n")
print(sort(abs(rs), decreasing = TRUE)[1:3])
cat("fitted at that row:", fitted(m1)[i], "\n")
m2 <- lm(footfall ~ sales + I(sales^2), df)
cat("R2 linear =", summary(m1)$r.squared, " R2 quad =", summary(m2)$r.squared,
    " adjR2 lin =", summary(m1)$adj.r.squared, " adjR2 quad =", summary(m2)$adj.r.squared,
"\n")
print(anova(m1, m2))
# without outlier
d2 <- df[-i, ]
cat("without outlier: r =", cor(d2$sales, d2$footfall), " slope =", coef(lm(footfall ~
sales, d2))[2], "\n")
df$flag <- seq_len(n) == i
p <- ggplot(df, aes(sales, footfall)) +
  geom_point(aes(colour = flag), size = 2.6) +
  geom_smooth(method = "lm", se = FALSE, colour = "#1f4e79") +
  geom_smooth(method = "loess", se = FALSE, colour = "#c0392b", linetype = "dashed") +
  geom_text_repel(data = df[i, ], aes(label = paste0("Row ", i, " (outlier)")), size = 3.2)
+
  scale_colour_manual(values = c("FALSE" = "grey30", "TRUE" = "#e67e22"), guide = "none") +
  labs(title = "Weekly sales vs footfall", subtitle = "Blue = straight line (lm), red dashed
= loess",
       x = "Weekly sales (Rs lakh)", y = "Footfall (visitors per day)") + th
print(p)    # show the chart in the Plots pane
ggsave("figs/q1.png", p, width = 6.5, height = 4, dpi = 200, bg = "white")
```

## Q2 Correlogram: telecom circle-periods

**Task 1** *Correlation matrix (2 decimals) and correlogram with diverging scale centred on 0.*

| | tower_density | drop_rate | traffic_tb | users_lakh | churn_pct | arpu |
|---|---|---|---|---|---|---|
| tower_density | 1.00 | -0.70 | 0.05 | -0.20 | -0.19 | -0.57 |
| drop_rate | -0.70 | 1.00 | 0.00 | 0.20 | 0.21 | 0.62 |
| traffic_tb | 0.05 | 0.00 | 1.00 | 0.78 | -0.13 | -0.05 |
| users_lakh | -0.20 | 0.20 | 0.78 | 1.00 | -0.04 | 0.14 |
| churn_pct | -0.19 | 0.21 | -0.13 | -0.04 | 1.00 | 0.02 |
| arpu | -0.57 | 0.62 | -0.05 | 0.14 | 0.02 | 1.00 |

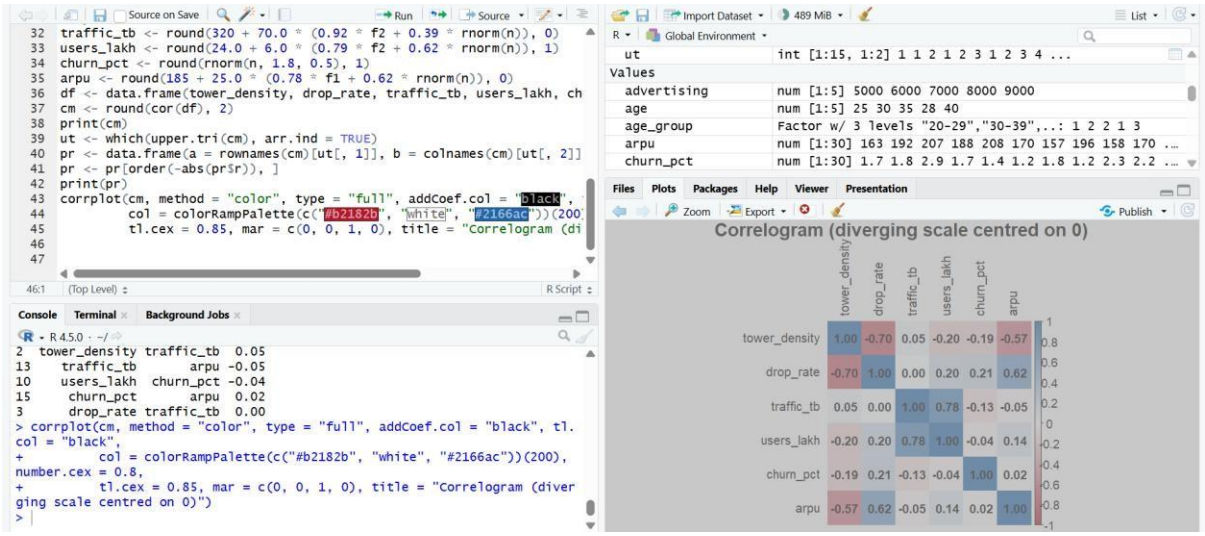



*Figure Q2: Correlogram of the six measures (blue = positive, red = negative; scale centred on 0).*

**Task 2** *Two most strongly correlated pairs.*

- **traffic_tb and users_lakh: r = 0.78** (circles with more users carry more data traffic).
- **tower_density and drop_rate: r = −0.70** (more towers per area means fewer dropped calls). The next strongest is drop_rate and arpu at 0.62.

**Task 3** *One surprising weak or negative correlation.*

**drop_rate and churn_pct: r = 0.21.** Dropped calls would be expected to push customers away, but the link to churn is weak, so call quality alone does not explain churn in these circles. (A related surprise: drop_rate and arpu are positively related at 0.62, which is counter-intuitive and most likely arises because both move with tower density, so it should not be read as cause and effect.)

**Task 4** *Recommendation for the Network Head.*

Prioritise **adding towers in low-density circles**: tower density has the strongest link to dropped calls (r = −0.70). Do not expect that fixing drop rate alone will cut churn (r = 0.21); look at other churn drivers such as pricing and competitor offers. With only 30 circle-periods these correlations are indicative rather than conclusive.

## R code used for Q2

```
library(corrplot)
dir.create("figs", showWarnings = FALSE)

set.seed(1296125)
n <- 30
f1 <- rnorm(n); f2 <- rnorm(n)
tower_density <- round(14.0 + 4.0 * (-0.90 * f1 + 0.44 * rnorm(n)), 1)
drop_rate <- round(2.6 + 0.7 * (0.88 * f1 + 0.47 * rnorm(n)), 1)
traffic_tb <- round(320 + 70.0 * (0.92 * f2 + 0.39 * rnorm(n)), 0)
users_lakh <- round(24.0 + 6.0 * (0.79 * f2 + 0.62 * rnorm(n)), 1)
churn_pct <- round(rnorm(n, 1.8, 0.5), 1)
arpu <- round(185 + 25.0 * (0.78 * f1 + 0.62 * rnorm(n)), 0)
df <- data.frame(tower_density, drop_rate, traffic_tb, users_lakh, churn_pct, arpu)
cm <- round(cor(df), 2)
print(cm)
ut <- which(upper.tri(cm), arr.ind = TRUE)
pr <- data.frame(a = rownames(cm)[ut[, 1]], b = colnames(cm)[ut[, 2]], r = cm[ut])
pr <- pr[order(-abs(pr$r)), ]
print(pr)
corrplot(cm, method = "color", type = "full", addCoef.col = "black", tl.col = "black",
         col = colorRampPalette(c("#b2182b", "white", "#2166ac"))(200), number.cex = 0.8,
         tl.cex = 0.85, mar = c(0, 0, 1, 0), title = "Correlogram (diverging scale centred
on 0)")
```

## Q3 Dimension reduction: PCA and t-SNE

**Task 1** *PCA with prcomp(df, scale. = TRUE).*

| | PC1 | PC2 | PC3 | PC4 | PC5 | PC6 |
|---|---|---|---|---|---|---|
| % variance | 34.4 | 20.8 | 18.0 | 11.4 | 9.6 | 5.8 |
| Cumulative % | 34.4 | 55.2 | 73.2 | 84.6 | 94.2 | 100.0 |

PC1 explains **34.4%** and PC2 explains **20.8%** of the variance (55.2% together). **Four components** are needed to reach at least 80% (84.6%); three give only 73.2%.

**Task 2** *PC1 vs PC2 coloured by grp; largest loading on PC1.*

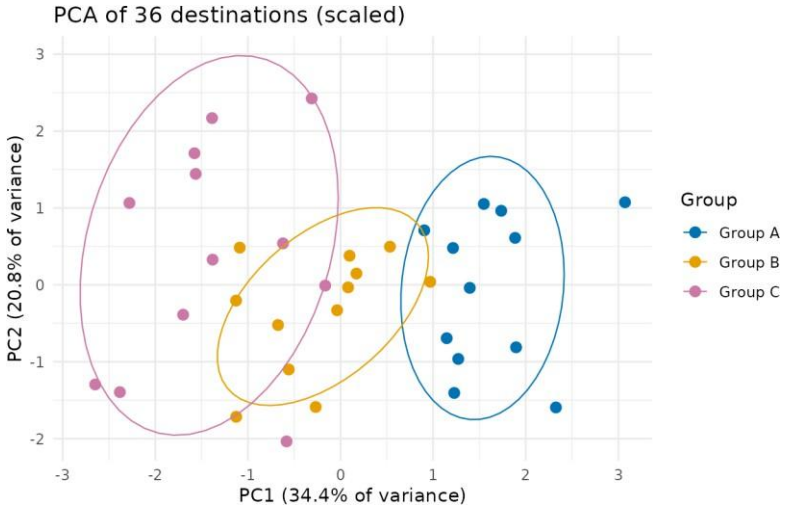



*Figure Q3a: PCA scores, PC1 vs PC2, coloured by group (ellipses cover 80% of each group).*

The variable with the largest absolute loading on PC1 is **complaints (0.590)**, closely followed by footfall_k (0.560) and rating10 (0.497). PC1 therefore behaves like a "size/busyness" axis. On PC2, stay_nights (0.670) and occupancy_pct (0.663) dominate.

**Task 3** *How well does PCA separate the groups?*

$R^2$ of lm(score ~ grp) is **0.772 for PC1** and **0.070 for PC2**. So 77% of the PC1 variation lies between groups, which means PC1 separates the three groups well (Group A on one side, Group C on the other, Group B in the middle), while PC2 carries almost no group information (7%). Group B overlaps with its neighbours, so separation is good but not perfect.

**Task 4** *t-SNE plot, trade-off, recommendation.*

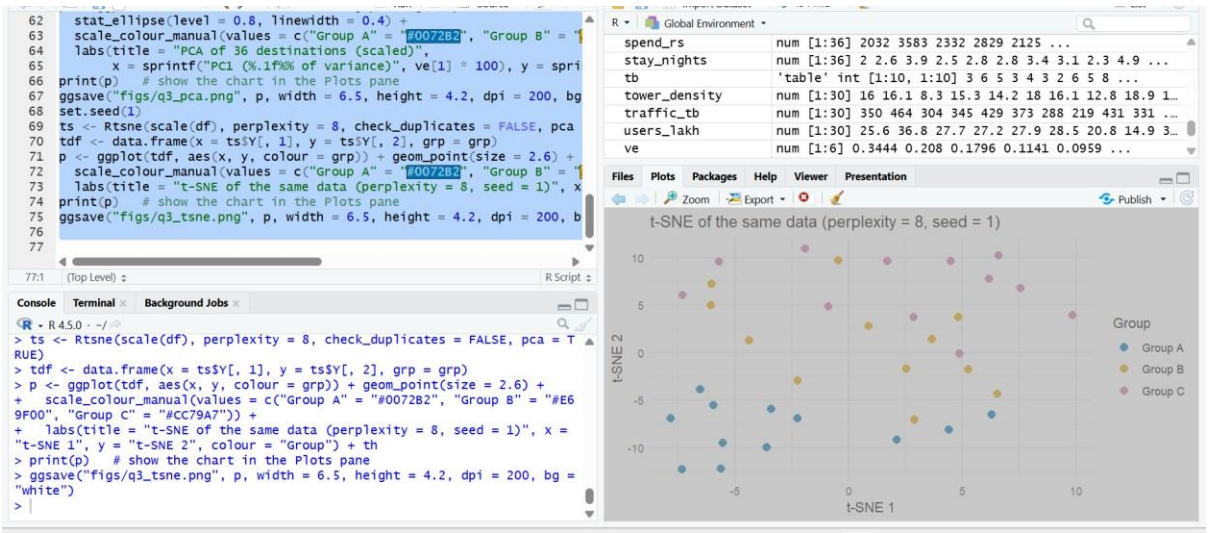



*Figure Q3b: t-SNE of the same scaled data (perplexity = 8, seed = 1).*

t-SNE (perplexity 8) shows the same ordering: Group A mostly on the left, Group C on the right and Group B mixed between them, with some overlap. **Trade-off:** PCA is linear and keeps the directions of largest **variance**, so global distances and the loadings are meaningful and easy to explain, but it can miss curved structure. t-SNE preserves **local neighbourhoods**, so it can reveal clusters that PCA flattens, but distances between clusters and the axes have no meaning, the picture changes with perplexity and seed, and it gives no loadings.

**Recommendation:** use the PC1 score as the main ranking/segmentation measure for destinations (it is explainable and separates the groups), and use t-SNE only as a visual cross-check, not as evidence of how far apart groups are.

### R code used for Q3

```
library(ggplot2);  library(Rtsne)
th <- theme_minimal(base_size = 11)
dir.create("figs", showWarnings = FALSE)

set.seed(1296182)
n <- 36
grp <- rep(c("Group C", "Group B", "Group A"), each = 12)
g <- rep(c(-1, 0, 1), each = 12)
footfall_k <- round(120.0 + 30.0 * (rnorm(n) + 0.90 * g), 1)
spend_rs <- round(3200 + 600.0 * (rnorm(n) + 0.20 * g), 0)
stay_nights <- round(2.8 + 0.7 * (rnorm(n) + 0.09 * g), 1)
rating10 <- round(7.6 + 0.8 * (rnorm(n) + 1.20 * g), 1)
occupancy_pct <- round(68.0 + 10.0 * (rnorm(n) + 0.06 * g), 1)
complaints <- round(4.0 + 1.2 * (rnorm(n) + 1.20 * g), 1)
df <- data.frame(footfall_k, spend_rs, stay_nights, rating10, occupancy_pct, complaints)
pc <- prcomp(df, scale. = TRUE)
print(summary(pc))
ve <- pc$sdev^2 / sum(pc$sdev^2)
cat("var explained:", round(ve * 100, 2), "\ncumulative:", round(cumsum(ve) * 100, 2), "\n")
print(round(pc$rotation[, 1:2], 3))
sc <- as.data.frame(pc$x); sc$grp <- grp
cat("R2 PC1~grp:", summary(lm(PC1 ~ grp, sc))$r.squared, " R2 PC2~grp:", summary(lm(PC2 ~
grp, sc))$r.squared, "\n")
p <- ggplot(sc, aes(PC1, PC2, colour = grp)) + geom_point(size = 2.6) +
  stat_ellipse(level = 0.8, linewidth = 0.4) +
  scale_colour_manual(values = c("Group A" = "#0072B2", "Group B" = "#E69F00", "Group C" =
"#CC79A7")) +
  labs(title = "PCA of 36 destinations (scaled)",
       x = sprintf("PC1 (%.1f%% of variance)", ve[1] * 100), y = sprintf("PC2 (%.1f%% of
variance)", ve[2] * 100), colour = "Group") + th
print(p)    # show the chart in the Plots pane
ggsave("figs/q3_pca.png", p, width = 6.5, height = 4.2, dpi = 200, bg = "white")
set.seed(1)
ts <- Rtsne(scale(df), perplexity = 8, check_duplicates = FALSE, pca = TRUE)
tdf <- data.frame(x = ts$Y[, 1], y = ts$Y[, 2], grp = grp)
p <- ggplot(tdf, aes(x, y, colour = grp)) + geom_point(size = 2.6) +
  scale_colour_manual(values = c("Group A" = "#0072B2", "Group B" = "#E69F00", "Group C" =
"#CC79A7")) +
  labs(title = "t-SNE of the same data (perplexity = 8, seed = 1)", x = "t-SNE 1", y = "t-
SNE 2", colour = "Group") + th
print(p)    # show the chart in the Plots pane
ggsave("figs/q3_tsne.png", p, width = 6.5, height = 4.2, dpi = 200, bg = "white")
```

## Q4 Network graph: parcel flows between 7 hubs (5 marks)

**Task 1** *Network with igraph: node size ∝ degree, edge width ∝ weight.*

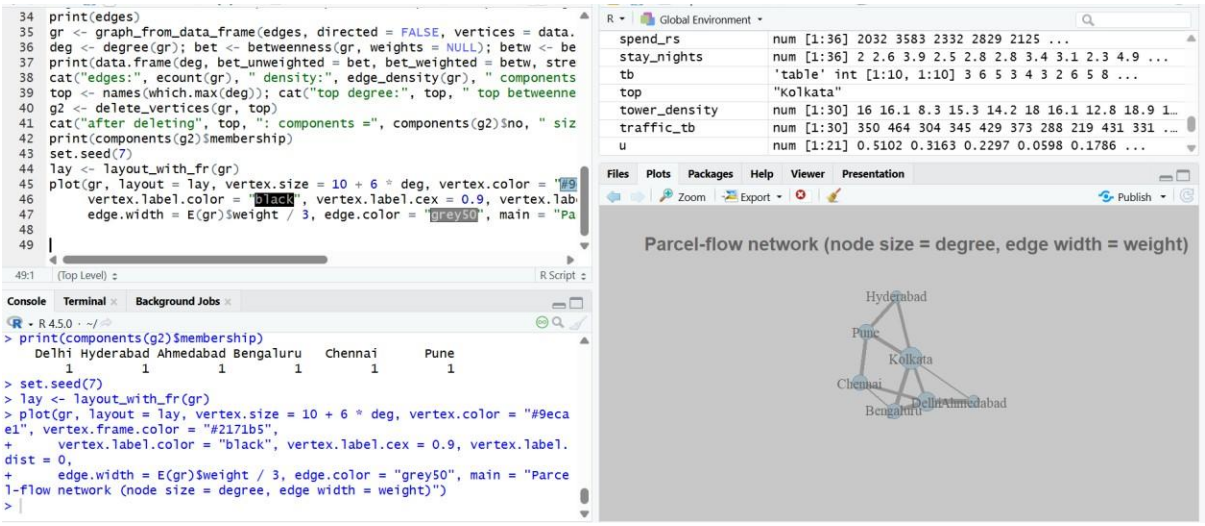



*Figure Q4: Parcel-flow network. Node size shows degree; edge width shows weight.*

**Task 2** *Degree and betweenness; most central node.*

| Hub | Degree | Betweenness | Strength (flow) | Note |
|---|---|---|---|---|
| Kolkata | 5 | 6.5 | 54 | Highest on all |
| Delhi | 4 | 0.0 | 52 | |
| Bengaluru | 3 | 2.0 | 35 | |
| Chennai | 3 | 2.5 | 28 | |
| Pune | 3 | 1.0 | 38 | |
| Hyderabad | 2 | 0.0 | 22 | |
| Ahmedabad | 2 | 0.0 | 21 | |

**Kolkata** is the most central node: it has the highest degree (**5** links) and the highest betweenness (**6.5**, compared with 2.5 for the next hub, Chennai). It connects directly to 5 of the other 6 hubs and sits on most shortest routes between the rest.

**Task 3** *Edges, density, components.*

**11 edges**; **density = 0.524** (11 of the 21 possible links); **1 connected component** (every hub can reach every other).

**Task 4** *Delete the most central node; recommendation.*

After deleting Kolkata the network still has **1 component** (the remaining 6 hubs stay connected), but several routes become longer and Hyderabad and Ahmedabad are left with a single link each.

**Recommendation for the Operations Head:** the network will not break if Kolkata is lost, but Kolkata carries the most routes and flow, so add spare capacity or a backup plan there, and add a direct link for Hyderabad and Ahmedabad (for example Hyderabad–Chennai) so they do not depend on a single route.

### R code used for Q4

```
library(igraph)
dir.create("figs", showWarnings = FALSE)

set.seed(1296212)
nodes <- c("Delhi", "Hyderabad", "Kolkata", "Ahmedabad", "Bengaluru", "Chennai", "Pune")
pairs <- t(combn(nodes, 2))
u <- runif(nrow(pairs))
```

```
keep <- u < 0.28 | ((pairs[, 1] == "Kolkata" | pairs[, 2] == "Kolkata") & u < 0.86)
edges <- data.frame(from = pairs[keep, 1], to = pairs[keep, 2], weight =
round(runif(sum(keep), 1, 20)))
print(edges)
gr <- graph_from_data_frame(edges, directed = FALSE, vertices = data.frame(name = nodes))
deg <- degree(gr); bet <- betweenness(gr, weights = NULL); betw <- betweenness(gr)
print(data.frame(deg, bet_unweighted = bet, bet_weighted = betw, strength = strength(gr)))
cat("edges:", ecount(gr), " density:", edge_density(gr), " components:", components(gr)$no,
"\n")
top <- names(which.max(deg)); cat("top degree:", top, " top betweenness:",
names(which.max(bet)), "\n")
g2 <- delete_vertices(gr, top)
cat("after deleting", top, ": components =", components(g2)$no, " sizes:",
components(g2)$csize, "\n")
print(components(g2)$membership)
set.seed(7)
lay <- layout_with_fr(gr)
plot(gr, layout = lay, vertex.size = 10 + 6 * deg, vertex.color = "#9ecae1",
vertex.frame.color = "#2171b5",
     vertex.label.color = "black", vertex.label.cex = 0.9, vertex.label.dist = 0,
     edge.width = E(gr)$weight / 3, edge.color = "grey50", main = "Parcel-flow network (node
size = degree, edge width = weight)")
```

## Q5 Paired / slope chart: wait time before vs after (5 marks)

**Task 1** *Slope chart with labelled endpoints.*

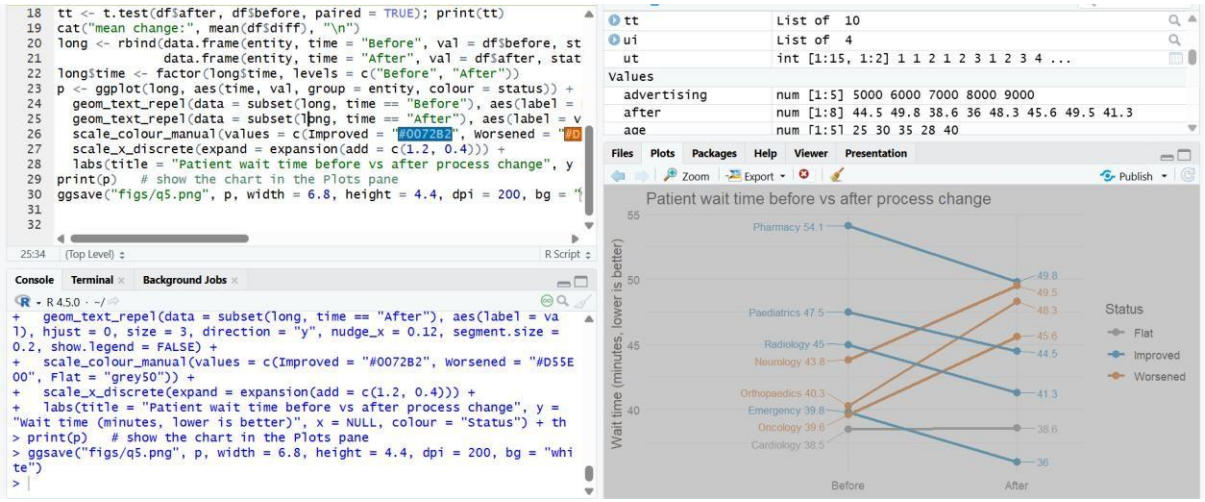



*Figure Q5: Patient wait time (minutes) by department before and after the process change (lower is better).*

**Task 2** *Classify improved / worsened / flat.*

Flat means an absolute change below 1% of the mean "before" value. The mean before is 43.6 minutes, so the cut-off is about 0.44 minutes (the sheet quotes 0.420, which is 1% of the nominal mean of 42.0; the classification is the same either way because the smallest other change is 3.0 minutes).

| Department | Before | After | Change | % of before | Status |
|---|---|---|---|---|---|
| Paediatrics | 47.5 | 44.5 | -3.0 | -6.3% | Improved |
| Pharmacy | 54.1 | 49.8 | -4.3 | -7.9% | Improved |
| Cardiology | 38.5 | 38.6 | +0.1 | +0.3% | Flat |
| Emergency | 39.8 | 36.0 | -3.8 | -9.5% | Improved |
| Orthopaedics | 40.3 | 48.3 | +8.0 | +19.9% | Worsened |
| Oncology | 39.6 | 45.6 | +6.0 | +15.2% | Worsened |
| Neurology | 43.8 | 49.5 | +5.7 | +13.0% | Worsened |
| Radiology | 45.0 | 41.3 | -3.7 | -8.2% | Improved |

**Counts: 4 improved** (Paediatrics, Pharmacy, Emergency, Radiology), **3 worsened** (Orthopaedics, Oncology, Neurology) and **1 flat** (Cardiology).

### Task 3 *Largest change.*

**Orthopaedics** changed the most: wait time rose by **8.0 minutes**, which is **19.9%** of its "before" value (40.3).

### Task 4 *Paired t-test and recommendation.*

t.test(after, before, paired = TRUE): **t = 0.344, df = 7, p = 0.741**, mean change = **+0.625 minutes** (95% CI −3.67 to +4.92). The average change is tiny and **not statistically significant**: the improvements and the deteriorations cancel out.

**Recommendation:** do not roll the new process out everywhere on this evidence. Investigate why Orthopaedics, Oncology and Neurology got 5.7–8.0 minutes slower, and copy what worked in Pharmacy and Emergency.

### R code used for Q5

```
library(ggplot2); library(ggrepel)
th <- theme_minimal(base_size = 11)
dir.create("figs", showWarnings = FALSE)

set.seed(1296294)
entity <- c("Paediatrics", "Pharmacy", "Cardiology", "Emergency", "Orthopaedics",
"Oncology", "Neurology", "Radiology")
n <- 8
before <- round(rnorm(n, 42.0, 5.0), 1)
change <- round(c(-1, -1, -1, -1, 1, 1, 1, -1) * runif(n, 0.4, 1.4) * 6.0, 1)
change[3] <- 0.1
after <- before + change
df <- data.frame(entity, before, after)
df$diff <- round(df$after - df$before, 1)
df$pct <- round(100 * df$diff / df$before, 1)
thr <- 0.01 * mean(df$before)
df$status <- ifelse(abs(df$diff) < thr, "Flat", ifelse(df$diff < 0, "Improved", "Worsened"))
print(df); cat("mean before:", mean(df$before), " threshold:", thr, "\n");
print(table(df$status))
tt <- t.test(df$after, df$before, paired = TRUE); print(tt)
cat("mean change:", mean(df$diff), "\n")
long <- rbind(data.frame(entity, time = "Before", val = df$before, status = df$status),
              data.frame(entity, time = "After", val = df$after, status = df$status))
long$time <- factor(long$time, levels = c("Before", "After"))
p <- ggplot(long, aes(time, val, group = entity, colour = status)) + geom_line(linewidth =
1) + geom_point(size = 2.5) +
  geom_text_repel(data = subset(long, time == "Before"), aes(label = paste0(entity, " ",
val)), hjust = 1, size = 3, direction = "y", nudge_x = -0.15, segment.size = 0.2,
show.legend = FALSE) +
  geom_text_repel(data = subset(long, time == "After"), aes(label = val), hjust = 0, size =
3, direction = "y", nudge_x = 0.12, segment.size = 0.2, show.legend = FALSE) +
  scale_colour_manual(values = c(Improved = "#0072B2", Worsened = "#D55E00", Flat =
"grey50")) +
  scale_x_discrete(expand = expansion(add = c(1.2, 0.4))) +
  labs(title = "Patient wait time before vs after process change", y = "Wait time (minutes,
lower is better)", x = NULL, colour = "Status") + th
print(p)    # show the chart in the Plots pane
ggsave("figs/q5.png", p, width = 6.8, height = 4.4, dpi = 200, bg = "white")
```

## Q6 Bubble chart: genre marketing spend, watch time, subscribers

**Task 1** *Bubble chart with size scaling and labels.*

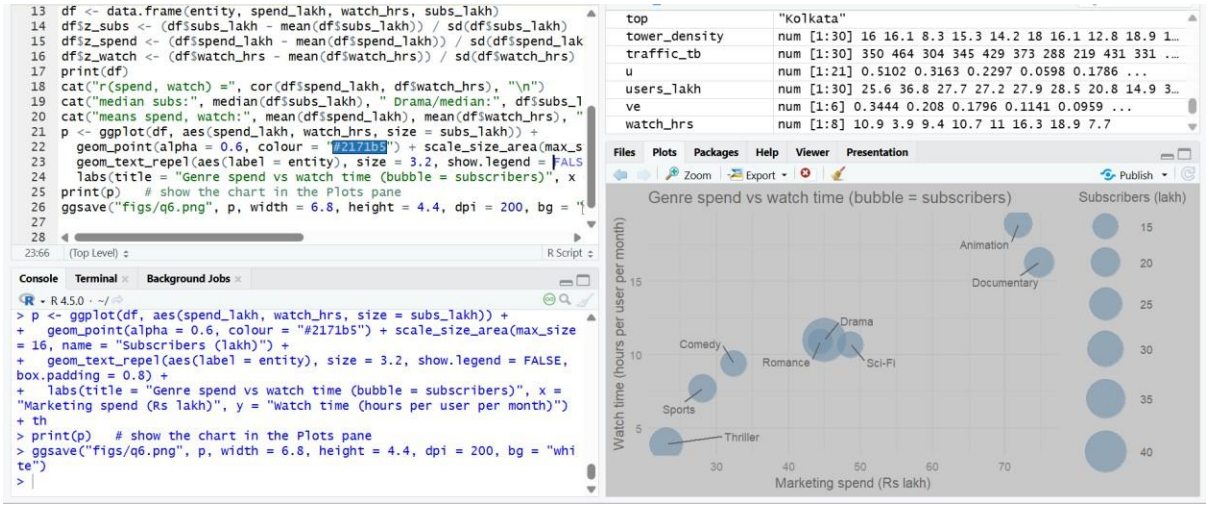



*Figure Q6: Marketing spend (Rs lakh) vs watch time (hours per user per month); bubble area = subscribers (lakh).*

**Task 2** *Correlation between x and y.*

Spend and watch time have **r = 0.96**, a very strong positive relationship: genres that receive more marketing spend also have far longer watch time per user.

**Task 3** *Outlier on the size variable.*

**Drama** is the entity. Its position is near average on both axes (spend 45.0 vs mean 46.0 Rs lakh; watch time 11.0 vs mean 11.1 hours) but its subscribers are **43.2 lakh**, a **z-score of +2.31** (mean 21.8, the next highest is Thriller at 25.3).

**Task 4** *Size relative to the median.*

The median subscribers is 18.5 lakh, so Drama is **about 2.3 times larger** (43.2 / 18.5 = 2.34). A plain scatterplot draws every genre as the same-sized dot and Drama would sit unnoticed among Romance and Sci-Fi in the middle of the trend line, so its huge subscriber base would be completely hidden.

**Task 5** *Recommendation for the Content Head.*

Drama gets about 0.96 lakh subscribers per lakh spent, against about 0.29 for Documentary and 0.25 for Animation, which have the highest spend and watch time but only average subscriber counts. **Protect and grow the Drama budget**, and review whether the heavy spend on Documentary and Animation is converting to subscribers.

### R code used for Q6

```
library(ggplot2); library(ggrepel)
th <- theme_minimal(base_size = 11)
dir.create("figs", showWarnings = FALSE)

set.seed(1296318)
entity <- c("Romance", "Thriller", "Comedy", "Sci-Fi", "Drama", "Documentary", "Animation",
"Sports")
n <- 8
z <- rnorm(n)
spend_lakh <- round(45.0 + 12.0 * z, 1)
watch_hrs <- round(11.0 + 3.0 * (0.90 * z + 0.40 * rnorm(n)), 1)
subs_lakh <- round(18.0 + 4.0 * rnorm(n), 1)
subs_lakh[5] <- round(18.0 * 2.4, 1); spend_lakh[5] <- round(45.0); watch_hrs[5] <-
round(11.0)
df <- data.frame(entity, spend_lakh, watch_hrs, subs_lakh)
df$z_subs <- (df$subs_lakh - mean(df$subs_lakh)) / sd(df$subs_lakh)
df$z_spend <- (df$spend_lakh - mean(df$spend_lakh)) / sd(df$spend_lakh)
df$z_watch <- (df$watch_hrs - mean(df$watch_hrs)) / sd(df$watch_hrs)
print(df)
cat("r(spend, watch) =", cor(df$spend_lakh, df$watch_hrs), "\n")
```

```
cat("median subs:", median(df$subs_lakh), " Drama/median:", df$subs_lakh[5] /
median(df$subs_lakh), "\n")
cat("means spend, watch:", mean(df$spend_lakh), mean(df$watch_hrs), "\n")
p <- ggplot(df, aes(spend_lakh, watch_hrs, size = subs_lakh)) +
  geom_point(alpha = 0.6, colour = "#2171b5") + scale_size_area(max_size = 16, name =
"Subscribers (lakh)") +
  geom_text_repel(aes(label = entity), size = 3.2, show.legend = FALSE, box.padding = 0.8) +
  labs(title = "Genre spend vs watch time (bubble = subscribers)", x = "Marketing spend (Rs
lakh)", y = "Watch time (hours per user per month)") + th
print(p)    # show the chart in the Plots pane
ggsave("figs/q6.png", p, width = 6.8, height = 4.4, dpi = 200, bg = "white")
```

## Q7 Hexbin plot: loan amount vs applicant income (5 marks)

**Task 1** *Hexbin plot with count legend and viridis scale, plus plain scatterplot.*

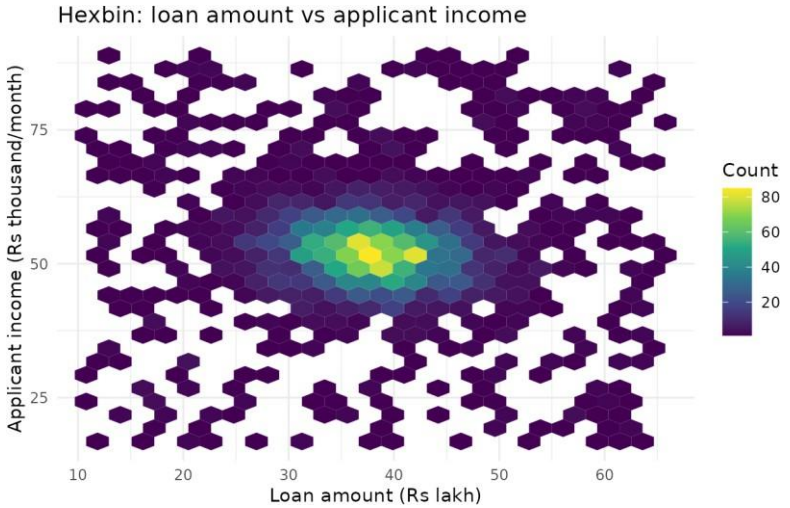



*Figure Q7a: Hexbin (bins = 25), loan amount (Rs lakh) vs applicant income (Rs thousand per month); colour = record count.*

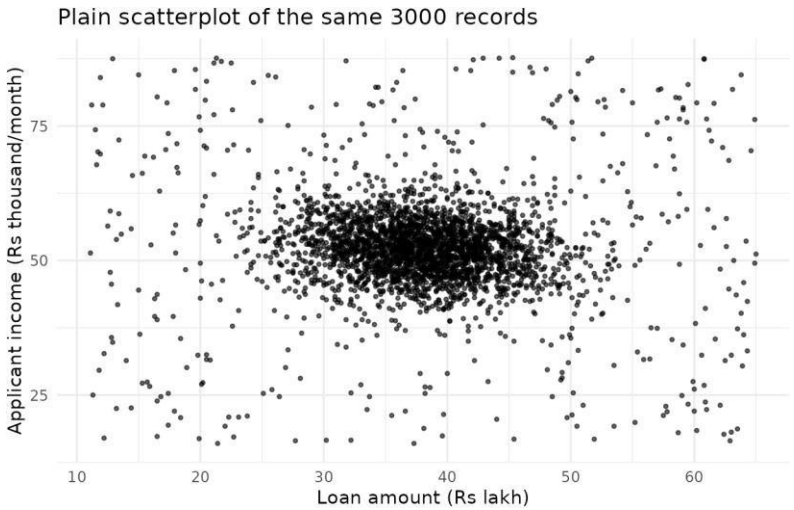



*Figure Q7b: Plain scatterplot of the same 3,000 records.*

**Task 2** *10 × 10 grid: busiest cell.*

Using cut(loan_lakh, 10) and cut(income_k, 10), the busiest cell is **loan (32.7, 38] Rs lakh** and **income (51.9, 59] Rs thousand per month**, containing **400 records = 13.3%** of all 3,000.

**Task 3** *Distinct (x, y) pairs.*

nrow(unique(df)) = **2,901** distinct pairs out of 3,000 rows, so 99 rows repeat an earlier pair exactly. In the plain scatterplot those repeats sit on top of each other, and thousands of points overlap in the dense centre, so the scatterplot **looks like one solid blob and badly**

**understates how concentrated the data is**; the 400-record cell looks no different from a cell with 40.

**Task 4** *Recommendation for the Zonal Manager.*

Most applicants fall in one core segment: loans of about **Rs 33–43 lakh** with income of about **Rs 45–59 thousand per month** (the four busiest cells hold 348–400 records each). Build the standard product, pricing and approval process around this segment, and send the remaining scattered applicants (very low or very high loans and incomes) to manual review.

### R code used for Q7

```
library(ggplot2); library(hexbin)
th <- theme_minimal(base_size = 11)
dir.create("figs", showWarnings = FALSE)

set.seed(1296371)
n <- 3000
core <- runif(n) < 0.85
loan_lakh <- round(ifelse(core, rnorm(n, 38.0, 5.4), runif(n, 11.0, 65.0)), 1)
income_k <- round(ifelse(core, 52.0 - 0.73 * (loan_lakh - 38.0) / 5.4 + sqrt(1 - 0.54) * 7.2
* rnorm(n), runif(n, 16.0, 88.0)), 1)
df <- data.frame(loan_lakh, income_k)
cat("r:", cor(df$loan_lakh, df$income_k), "\n")
gx <- cut(df$loan_lakh, 10); gy <- cut(df$income_k, 10)
tb <- table(gx, gy)
mx <- which(tb == max(tb), arr.ind = TRUE)
cat("busiest cell:", levels(gx)[mx[1, 1]], levels(gy)[mx[1, 2]], " count:", max(tb), " share
%:", 100 * max(tb) / n, "\n")
cat("cell counts matrix:\n"); print(tb)
cat("n rows:", nrow(df), " distinct (x,y):", nrow(unique(df)), "\n")
cat("range x:", range(df$loan_lakh), " range y:", range(df$income_k), "\n")
cat("overall busiest cell via cut(): ", max(tb), "\n")
p <- ggplot(df, aes(loan_lakh, income_k)) + geom_hex(bins = 25) + scale_fill_viridis_c(name
= "Count") +
  labs(title = "Hexbin: loan amount vs applicant income", x = "Loan amount (Rs lakh)", y =
"Applicant income (Rs thousand/month)") + th
print(p)    # show the chart in the Plots pane
ggsave("figs/q7_hex.png", p, width = 6.5, height = 4.2, dpi = 200, bg = "white")
p <- ggplot(df, aes(loan_lakh, income_k)) + geom_point(alpha = 0.6, size = 0.8) +
  labs(title = "Plain scatterplot of the same 3000 records", x = "Loan amount (Rs lakh)", y
= "Applicant income (Rs thousand/month)") + th
print(p)    # show the chart in the Plots pane
ggsave("figs/q7 scatter.png", p, width = 6.5, height = 4.2, dpi = 200, bg = "white")
```